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Technical drawing course, video games, gender, and type of school on spatial ability

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ABSTRACT

The authors investigated the effects of a technical drawing course, video games, gender, and type of school on the spatial ability (spatial visualization and orientation) of secondary school students in Barbados. A total of 420 students (269 boys; 151 girls) from nine government schools within Category 3 and Category 4 were sampled. A four-way between-subjects analysis of covariance revealed significant main effects for technical drawing on spatial visualization and orientation and video games on spatial orientation. Significant two- and three-way interactions occurred, but four-way interactions failed to reach significance. Implications are discussed for secondary school teachers and educational stakeholders on improving practice and context through the innovative delivery of spatial content, with consideration of biological and social factors that affect spatial abilities.

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For several decades, there has been considerable debate regarding the improvement of programs that aid the development of the spatial skill of students, especially for those exiting school for related occupations. The government of Caribbean states understood the importance of spatial abilities to succeed, and the role of technical and vocational subjects. These states directed the Caribbean Examination Council to provide examinations that assessed the students that were leaving school for educational advancement or entry to work. Technical and vocational areas such as technical drawing (TD) were developed and introduced as industrial arts in secondary schools (Caribbean Examination Council, 2000). Every student in secondary school should acquire fundamental knowledge of TD and other technical and vocational areas.

The Barbados curriculum reform in 2000 within primary and secondary schools focused on an outcome-based educational approach to increase student competency in different subject areas (Ministry of Education, Youth Affairs and Culture, 2000; Planning and Research Section, 2000). Increases in spatial and technological experiences in the vocational and academic classrooms occurred, as compared with the previous educational focus.

Secondary schools are differentiated by ranking based on the national Barbados Secondary School Entrance Examination (BSSEE) academic performance. These categories of school are based on students BSSEE average of the English and mathematics scores. Within this study, the type of school referred to Category 1, Category 2, Category 3, and Category 4 coeducational government secondary schools as classified by Leacock, Thompson, Burnett, and Obidah (2007). Category 3 and Category 4 schools were targeted because there were naturally occurring comparison groups, as some schools did not expose students to TD. Also, most schools in Category 1 and Category 2 did not want to participate in the study. A final reason for the study of these categories was that a significant number of the students in Category 3 and Category 4 schools often transitioned from school to work at an earlier age, and the fundamental knowledge of TD may be more beneficial. In Barbadian secondary schools, second- and third-form students follow a traditional academic program of study, which includes industrial arts or vocational representation (Ministry of Education, Youth Affairs and Culture, 1995; Ministry of Education, Youth Affairs and Culture, 2000; Planning and Research Section, 2000). At the third form level, students are allowed to select some grouped subjects, which are predetermined by the school. There are instances in which students entering the third form level opt not to continue with the industrial arts or vocational programs due to a focus on academic areas. However, sometimes students are partially exposed to vocational courses, as schools may offer hybrid programs with options of teaching TD for six weeks. In other schools, there is the decision to eliminate the TD course and related courses. Therefore, we pondered if this basic course being taught in secondary schools at the third form level (approximately 14 years old) would cause a significant difference in the spatial ability of students. This study is needed at this time because it could aid in a major curriculum reform, which has not occurred since the year 2000, and highlight innovative instructional strategies and gender differences regarding spatial performances.

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The literature illustrated the importance of spatial ability for successful endeavors in courses related to engineering and technical education (Bartholomew, 2014; Burton & Dowling, 2010; McCuen, 2015). Spatial ability is defined as having components that involve spatial visualization and spatial orientation (McGee, 1979). While spatial visualization occurs when an observer mentally manipulates two- and three-dimensional objects, spatial orientation is the observer mentally manipulating his or her body to view two- and three-dimensional objects. Students were instructed in the TD classes, where various concepts such as projections of isometric and orthographic were heavily linked to different spatial abilities and development of spatial skills such as rotation and changing looking directions (Serdar & Harm de Vries, 2015). Major features of these classes demand students to mentally manipulate and visualize several items and conditions, which may facilitate successful performances within aspects of their lives as they encounter opportunities to assess and solve spatial tasks. These students must produce drawings in which they have to spatially visualize and orientate objects and themselves in a manner that is efficient and effective. Related occupational fields such as building, civil engineering, farming, and architecture will require competence in such skills.

The National Research Council (2006) highlighted a situation in which the spatial cognitive skills of students were not cultivated due to opportunities not being in abundance. Sorby, Wysocki, and Baartmans (2003) offered the norm of structuring educational programs facilitates reduction of precise training in spatial ability. This departure from industrial arts and vocational programs could be the result of parents and teachers who consider spatial activities such as TD, metalwork, woodwork, and so forth as nonacademic subjects. Spruyt, Van Droogenbroeck, and Kavadias (2015) examined documentation from a study in Flanders regarding stigma concerning perceptions about technical and vocational education in secondary school. It was concluded that technical and vocational education students perceived that general education was more approved than the technical and vocational education.

Several studies offered differing spatial factor definitions and structures. Olkun (2003) purported spatial ability involved concepts associated with mental operations of objects in a two- and three-dimensional space. McGee (1979) indicated the actuality of visualization and orientation as two spatial factors as derived from factor analytic and predictive validity studies. Spatial visualization was the observer mentally manipulating of objects, and spatial orientation was mental manipulation of the observer's body to view objects. Similarly, Risma (2013) and Cakmak, Isiksal, and Koc (2014) agreed that spatial visualization and spatial orientation were components of spatial ability. It is recognized that spatial visualization is a broad construct with several elements, but in the present study, mental rotation will be the focus. Targeting of mental rotation is deliberate because students at the third form level perform related tasks. Uttal, Meadow, et al. (2013) advanced an alternative perspective with their classification of spatial tasks concerning a distinction of information. The first distinction of information was between intrinsic (comparing an object's parts) and extrinsic (comparing objects within their surrounding environment). The other distinction was between dynamic (advance cognitive interaction with an object) and static (recognition of an object). TD tasks for the students can be considered intrinsic-dynamic information for spatial visualization and extrinsic-static information for spatial orientation.

Investigations into spatial development are highlighted at varying ages and academic levels (De Lisi & Wolford, 2002; Kayhan, 2005; Sorby, Casey, Veurink, & Dulaney, 2013). Much of these investigations demonstrate that spatial ability can be improved through instruction. Sorby et al. (2013) assessed the mental rotation of first-year students assigned to spatial training and control groups. Results suggested the trained group improved greater than they would have without training, and there was transferred learning extended to calculus. Uttal, Meadow, et al. (2013) and Uttal, Miller, and Newcombe (2013) proposed that spatial training and life experiences can improve different spatial abilities. Spatial training can be informal such as playing video games or formal such as taking science, technology, engineering, and mathematics (STEM) subjects.

There has been a significant role attributed to spatial ability in the human capacity as individuals interact, organize, and observe their environment (McGee, 1979). McCuen (2015) offered that different spatial abilities were fundamental to human success in many areas, such as academic studies, domestic activities, and various occupations. However, the level of innate spatial ability was not equal in all human beings; hence, strategies to develop abilities are a priority of the educational system (Kayhan, 2005; Marunic & Glazar, 2013; Sorby & Veurink, 2010). Spatial training such as TD provides students with learning tasks that specifically and consistently focus on developing spatial ability. Perkins, Hazelton, Erickson, and Allan (2010) defended schools' using a curriculum that encouraged developing students' awareness of their spatial ability. Sharobeam (2016) concluded that student spatial performance was higher in schools with an STEM program than in schools without STEM. However, school administrators, parents, and nonvocational teachers are still hesitant, although there is evidence about the importance of such training to enhance spatial ability.

Research in electronics suggested experiences with video games and computer use can facilitate spatial ability development in several academic and vocational areas (David, 2012; Motamedi & Yaghoubi, 2015). De Castell, Larios, Jenson, and Smith (2015) proposed a positive relationship between spatial task performances and individuals exposed to previous activities involving the manipulation of objects and video games. Gagnon (1985) discovered a correlation between spatial tests scores and video game scores, with experienced video game players having higher spatial test scores. Gold et al. (2018) offered that boys' early video game experiences allow for developing spatial skills that create an advantage over girls. Feng, Spence, and Pratt (2007)

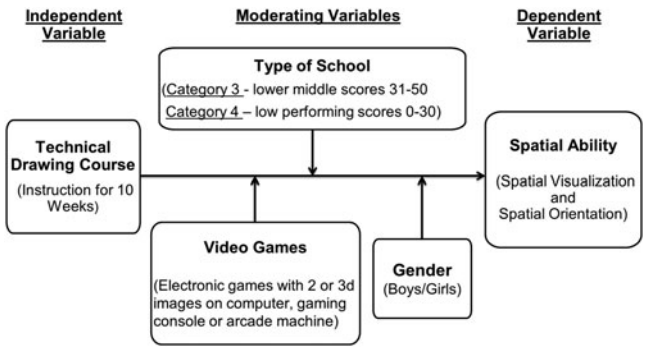

Figure 1. Conceptual framework of the effect of a technical drawing course on spatial ability with moderating variables video games, gender, and type of school. Spatial ability components are spatial visualization and spatial orientation (McGee, 1979; Risma, 2013).

suggested that three-dimensional action video games could remove gender differences in mental rotation ability. Similar research hypothesized the benefits of video games to enhance social, motivation and emotional development (Granic, Lobel, & Engels, 2014; Green & Bavelier, 2015; Uttal, Meadow, et al., 2013; Uttal, Miller, & Newcombe, 2013). Research on gender differences and video games showed that boys outperformed girls and vice versa (Cherney, Bersted, & Smetter, 2014; Spence & Feng, 2010; Terlecki & Newcombe, 2005). Spatial performance is affected by video games, which is significant for the present study. The prevalence of video games in this technological era and students' free play can be used to aid formal classroom instruction; hence, it should be considered within the present study.

Differences in the spatial performance of boys and girls is attributed to several proposed phenomena. Variations in spatial performance can be explained by various types of test, task, or personal encounters (Maeda & Yoon, 2013; Neuburger, Jansen, Heil, & Quaiser-Pohl, 2011); the interaction of social and biological factors (Bock & Kolakowski, 1973; Yazici, 2013); the type and amount of sex hormones present (Kimura, 1999); or the dominant use of brain hemispheres (Capraro, 2001; Safhalter, Glodez, & Vukman, 2013; Spencer-Smith et al., 2013; Thomason, Race, Burrows, Whitfield-Gabrieli, Glover, & Gabrieli, 2009) Other researchers have considered alternative explanations of gender differences such as genetics (O'Connor, 1943; Yazici, 2013), and prior spatial experiences (Alias, Black, & Gray, 2002; Baenninger & Newcombe, 1989; De Castell, Larios, Jenson, & Smith, 2015).

Conceptual framework

Extant related spatial literature facilitated the conceptual framework (see Figure 1) included in our present study. We considered the effects of a TD course, video games, gender, and the type of school on spatial ability, namely the constructs of spatial visualization and orientation.

TD course

Various studies have examined the effects of formal spatial training courses such as TD that involved physical and mental manipulation of figures. These studies have demonstrated the benefit of spatial training courses at improving spatial performance. In the conceptual framework within the present study, TD is an independent variable which affects the TD students' spatial ability.

According to Kayhan (2005), there is a link between TD and engineering drawing, in which both included visual skills and graphical communication. Kayhan investigated how a TD course affected the development of spatial visualization ability and spatial orientation ability. Data were collected from 221 Turkish ninth-grade students and results indicated that the TD course aided the spatial ability of students to develop significantly. Also, there was a significantly positive relationship between TD achievement and spatial ability. Student success in technical and vocational programs was linked to high spatial ability (Serdar & Harm de Vries, 2015). Lienhard (2000) agreed and reflected on his childhood and the importance of different spatial abilities to succeed in a vast number of vocational subjects. Although spatial intellect can be seen as relevant to TD students, it is not explicitly highlighted in the literature; however, there are studies that offer discussions with a focus on engineering students. Koch and Sanders (2011) indicated that spatial visualization ability was needed for technical problem-solving success; hence, teachers needed to include instruction and activity that teach basic visualization and problem-solving skills. Sorby and Baartmans (2000) developed a pre-graphics course in which they assessed 535 (418 boys, 117 girls) freshman engineering students with weak spatial visualization skills. The engineering students in subject areas related to TD were assessed using several tests, with the Vandenberg and Kuse (1978) Mental Rotations Test (MRT) relevant to our study. Results revealed significantly increased performance on the tests of students that completed the pre-graphics course. The students that completed the pre-graphics course were more likely to continue in engineering, and they outperformed students in the control groups. Sorby and Veurink (2010) recognized the value of a graphics course in improving the different spatial abilities of new engineering students, while Verner and Merksamer (2015) recommended that visual literacy can be developed in graphics courses.

Alias, Black, and Gray (2002) used a quasi-experimental design with pretest and posttest at Malaysian polytechnics with civil engineering students. They investigated the impact of classroom sketching and manipulating activities on the spatial visualization of students. Results showed a positive effect on spatial visualization ability in the classroom environment. It was discussed that spatial visualization had a significant impact on academic engineering success of students, as it was linked to other related subjects. Similarly, Katsioloudis, Jovanovic, and Jones (2014) examined the development of engineering students' spatial ability through a comparison of two- and three-dimensional drafting models. Students viewed various model types such as two-dimensional paper drawings, three-dimensional computer paper prints, and three-dimensional computer model printouts. These students were assessed using the MRT developed by Vandenberg and Kuse (1978). Katsioloudis, Jovanovic, and

Jones (2014) concluded that all methods were beneficial, but students exposed to three-dimensional computer images and computer-generated models outperformed those exposed to two-dimensional drawings.

Olkun's (2003) research gave middle-grade students activities that involved engineering drawing applications to improve their spatial ability. Engineering drawing was selected because of its practical demonstration of skills, and activities that both help to improve spatial abilities. Maier (1996) and Thurstone (1938) promoted the importance of spatial training at a specific age and the length of the training. After 14 years old, the development in spatial visualization ability would become stationary, although some growth in mental abilities would occur through the 20s. Uttal, Meadow, et al. (2013) and Uttal, Miller, and Newcombe (2013) offered spatial skills and thinking can be improved with specific training within all ages. Basham and Kotrlik (2008) proposed that the development of different spatial abilities in ninth-grade students is achievable if suitable training resources are present. The training resources included three-dimensional computer-aided design and drafting software and teacher-developed lesson plans used within a technology discovery course.

Video games

Electronics research in the literature indicated experiences with video games and using computers assisted the development of spatial ability. This experience is noteworthy because students recreationally play video games that may have positive effects on their spatial performance. However, students' pastime video game use is performed in uncontrolled informal settings. One should consider and investigate the effects of video game use on spatial performance. Therefore the conceptual framework posits that video games alter the effect of the TD course on students' spatial ability.

De Castell, Larios, Jenson, and Smith (2015) investigated how having experiences with video games may affect cognitive ability and academic success in STEM areas. Eighty-two students categorized as a nonexpert and expert players were assessed on their mental rotation and the path they took for several tasks of a game (virtual Morris water maze). The results showed that experts outperformed nonexperts regarding the length of the path traveled, and the expert players' mental rotation ability was significantly different from the nonexperts. The researchers noted there was an uncertainty of the relationship between video game performance and development in STEM areas. Motamedi and Yaghoubi (2015) examined the spatial ability of 300 male high school students concerning computer video game use. Findings indicated significant positive relationships between different spatial abilities (mental rotation and spatial visualization) and computer game use. David (2012) explored training in computer video games to improve different spatial abilities (mental rotation, spatial orientation, and spatial visualization) of 286 students. Results suggested that student spatial performance improved once the tasks of both the computer video game and spatial test were related, but spatial visualization had greater complexities, so computer video games may not have been the most effective way to train students. Peters, Chisholm, and Laeng (1995) found that video games did not affect spatial performance and suggested any change in spatial performance may be the result of video games and spatial tests having natural abilities.

Feng, Spence, and Pratt (2007) researched video games and education with the focus of spatial development of students. Their research centered on persons playing three-dimensional video games that simulate real life and the findings suggested action video games could remove gender differences in mental rotation ability. Men seem not to benefit as much as women when training occurred on action video games. Much of this type of research hypothesized the benefits of video games to enhance spatial and general cognitive development, motivation, social, and emotions of persons (Granic, Lobel, & Engels, 2014; Green & Bavelier, 2015; Spence & Feng, 2010; Uttal, Meadow, et al., 2013; Uttal, Miller, & Newcombe, 2013).

Cherney, Bersted, and Smetter (2014) studied gender differences with a focus on video game training using Nintendo Wii and GameCube consoles to increase mental rotation performance for 30 women and 30 men. They controlled for masculine and spatial activities experienced in childhood, which may increase gender differences when assessing spatial ability. Their results indicated that men significantly outperformed women on the MRT pretest, but there was no difference after the 1 hr of video game training, and trained women and men outperformed those not exposed to video game training. A conclusion suggested women seemed to benefit more from the training experience than men, as a ceiling effect may have been present for men.

Gold et al. (2018) concluded that boys have an advantage over girls because of their childhood experiences that develop their spatial skills. Boys are typically socialized to participate in activities associated with being male, such as playing video games. An opportunity to reduce any gender gap and disparity in spatial skill performance would be to allow the early engagement of girls with traditional male activities. McClurg and Chaille (1987) experimental research explored the impact of playing specifically chosen computer games on the spatial skills of fifth-, seventh-, and ninth-grade students. In the analyses of covariance, there was a significant treatment effect. However, there were no significant main effects or interaction discovered for gender or grade level. Both female and male students appeared to benefit from playing computer games, as their mental rotation ability improved.

De Lisi and Wolford (2002) observed the impact of playing computer games on the mental rotation of primary school third-grade students. A two-dimensional MRT was given to 47 students before and after playing computer games. An experimental group practiced mental rotation skills for a period of 11 sessions lasting 30 min. Both experimental and control groups were equal in performance on the mental rotation pretest, but the control group had lower posttest scores than the experimental group did. All children

with initial low mental rotation performance improved. The findings allude that teachers can implement classroom activities involving computer games within the instruction to increase mental rotation ability. Gagnon (1985) explored eye-hand coordination, spatial skills, and video game relationship for 58 graduate and undergraduate students. Experimental and control groups were randomly selected to practice on two video games in which the player is a shooter. Findings indicated a correlation was present between several spatial tests and video games scores. Initially, persons less experienced in playing video games improved, with the opposite effect occurring in persons with more significant experience. However, persons indicating that they previously played video games for a long time had the highest spatial tests and video games scores.

Prior experiences in the video game and computer video game use that involve spatial activities can affect different spatial abilities. It has been shown that there is a positive relationship between spatial task performances and individuals exposed to previous activities involving manipulation of objects, apparatus, and video games (Alias, Black, & Gray, 2002; Baenninger & Newcombe, 1989; De Castell, Larios, Jenson, & Smith, 2015).

Gender

The differences between the spatial performance of boys and girls have been highlighted in the literature. How boys and girls solve spatial problems can be affected by their preferred strategies, prior experiences, and hormones. This difference is vital to the study, as TD teachers use several instructional strategies to improve spatial performance, but which may create advantages for a particular gender. As such, gender should be as a moderator variable that alters how TD affects students' spatial ability.

There are various attributions to gender differences in spatial ability, where both girls and boys may spatially outperform each other (Hubona & Shirah, 2006; Linn & Petersen, 1985; McGee, 1979; Tartre, 1990) or there were no differences in their performance (Sanz de Acedo Lizarraga & Ganuza, 2003). Other contributions to variation in spatial ability, are types of test or tasks used for assessing spatial ability (Bodner & Guay, 1997; Linn & Petersen, 1985; Maeda & Yoon, 2013; Neuburger, Jansen, Heil, & Quaiser-Pohl, 2011).

Bock and Kolakowski (1973) implied that social and biological factors interact in complicated ways to result in gender differences. Yazici (2013) suggested that interest, past experiences, cognitive style, environment, and lifestyle were factors causing this spatial difference in individuals. Kimura (1999) advanced low levels of androgen and estrogen hormones affected boys and girls differently regarding spatial performance. Boys performed better on spatial tests when androgen levels were low and girls performed better with low estrogen levels in their menstrual cycle. Due to the previously mentioned results of students' menstrual cycle effect on spatial performance, it is relevant to the present study.

The lateralization theory proposed that verbal processing occurred in the left hemisphere, and this hemisphere matured faster in girls; in boys, the right hemisphere matured faster, giving prominence to nonverbal processing such as spatial performance (Capraro, 2001; Thomason et al., 2009). Therefore boys may outperform girls on spatial tasks due to hemispheric purpose matching (McGlone, 1980; Safhalter, Glodez, & Vukman, 2013; Spencer-Smith et al., 2013). Hence, boys could have an advantage regarding how they processed the nonverbal spatial tests used in this present study. Also, teachers possibly used activities or instructional methods that favored boys.

These theories are relevant to this present study as they indicate why certain performances on spatial instruments may have occurred in male and female students. One can further consider the importance of these biological and social factors affecting the effectiveness and efficiency of how teachers instruct. Also, the importance of student performance on the MRT developed by Vandenberg and Kuse (1978) and Spatial Orientation Test (SOT) developed by Hegarty and Waller (2004) and Kozhevnikov and Hegarty (2001).

Type of school

Researchers have investigated the use and importance of spatial ability within educational environments. What can be concluded is that individuals with high spatial ability have success in spatially related areas. However, spatial skills are infrequently integrated into classroom instruction and devalued in educational contexts. The type of school that allows for practices that help students improve spatial skills is significant. Hence, the effects of the school environment on spatial ability were considered.

Kayhan (2005) studied how the type of high school affected the development of spatial visualization ability and spatial orientation ability. Data were collected from 221 Turkish ninth-grade students (51 girls and 170 boys) in eight classrooms within five different high schools. Results revealed that the type of school had no significant effect on the development of students' spatial ability. Perkins, Hazelton, Erickson, and Allan (2010) suggested that students aware of their different spatial abilities can be developed when a spatially technological based curriculum and tasks were employed. Conditions to increase their awareness was linked to how relevant it was to student academic setting or school environment such as classroom and features around the school site. Guzel and Sener (2009) examined 145 Turkish students' spatial ability in two Anatolian high schools and two public high schools. Their findings illustrated higher spatial ability in geometry the achievement of students in Anatolian schools than in public schools. Sharobeam (2016) investigated greater than 850 college students' spatial visualization ability. Sharobeam examined if there were variances between students exposed to STEM subjects and non-STEM students. Findings showed that non-STEM students' spatial visualization skills were significantly lower than those of STEM students.

A TD course can be utilized as an effective way to improve students' spatial ability, as it facilitates training of mental manipulation and visualization of two- or three-dimensional objects items, and conditions that will be encountered in life. While Barbadian third-form students (13–15 years old) are exposed to subjects (woodwork, metalwork, and technology education) that may contain indirect spatial activities, these subjects do not specifically and consistently focus on training to the develop the spatial ability, the effects of these subjects were not determined. Although several studies on different spatial abilities exist, the present study is novel, especially in the Caribbean context. There are no known existing studies that have explored the effects of these combinations of variables on the spatial ability of secondary school students in Barbados and other Caribbean states. Uttal, Meadow, et al. (2013) reported the characteristics of 206 spatial studies in their meta-analysis, with no studies located within the Caribbean. The study provides a valuable contribution to spatial literature and learning environments research, which examines the variables that affect spatial ability. Thus, the purpose of this present research was to determine the effect of a TD course, video games, gender, and type of school on third-form students' spatial ability, namely spatial orientation and spatial visualization (mental rotation). The following research questions were formulated to guide this study:

Research Question 1: Is there a statistically significant main effect of a TD course, video games, gender, and type of school on third-form students' spatial ability?;

Research Question 2: Is there a statistically significant two-way interaction effect for all possible combinations of a TD course, video games, gender, and type of school on third-form students' spatial ability?;

Research Question 3: Is there a statistically significant three-way interaction effect for all possible combinations of a TD course, video games, gender, and type of school on third-form students' spatial ability?

Research Question 4: Is there a statistically significant four-way interaction effect for all possible combinations of a TD course, video games, gender, and type of school on third-form students' spatial ability?

Method

A quantitative methodology was used to investigate the effects of a TD course, video games, gender, and type of school on the spatial ability of secondary school students in Barbados. The study employed a $2 \times 2 \times 2 \times 2$ nonrandomized quasi-experimental pretest-posttest control group factorial design. This design allowed groups to be pretested before the TD instruction, then students were exposed to TD and posttested after instruction. Shadish, Cook, and Campbell (2002) showed that a quasi-experimental design was appropriate and suitable if a researcher may not be able to control random group assignments or cannot manipulate an independent variable. In an educational context, such a design is more difficult for a researcher to create artificial groupings for true experiments (Creswell, 2012). In Barbados, the nine coeducational schools within Categories 3 and 4 had a natural grouping of either having a TD course or not. Three of the four schools had a TD course in Category 3, and three of the five schools were exposed to the same course in Category 4. Schools were differentiated by rank based on students' scores of their English and mathematics average on the national BSSEE academic performance (Leacock, Thompson, Burnett, & Obidah, 2007). The schools were ranked as Category 1 (high performing) scores 71–100, Category 2 (upper middle) scores 51–70, Category 3 (lower middle) scores 31–50, and Category 4 (low performing) scores 0–30.

Table 1. Numbers for type of school with and without TD.

Type of School	n
Category 3 schools	
With TD	3
Male students	86
Female students	4
Without TD	1
Male students	66
Female students	45
Category 4 schools	
With TD	3
Male students	37
Female students	52
Without TD	2
Male students	80
Female students	50

TD = technical drawing.

Participants

Data were gathered utilizing purposive sampling at nine public secondary schools in Barbados to include all students within the third-form group (13–15 years old) from lower- to upper-class families. The selected schools were naturally organized, in which 179 students were exposed to TD and 241 were not exposed (see Table 1). There were 420 black students (269 boys and 151 girls); 201 students in Category 3 and 219 students in Category 4. The trained TD teachers (35–50 years old) had a minimum teaching experience of 10 years. These male teachers had also worked on correcting regional TD examinations, in which they were exposed to creating and validating items, and moderator training. It is acknowledged that male teachers could impact the results, as they are socialized differently than girls are and may utilize instructional strategies and activities that are aligned with their gender identity. The female students may be disadvantaged because of instructional strategies and activities that involve processing styles that favor boys; hence, biases toward a specific gender could result within the classroom. Florack (2012) offered that an adverse effect of student performance may occur if teachers created relationship gaps based on the social background and gender of students. There must be close monitoring of equality in classroom instruction and activities to ensure an unbiased teaching-learning environment.

Before the third form level, male and female students in these selected schools would have completed two years in a

technical and vocational program but would not have been exposed to TD. Male students would have been exposed to industrial arts with subjects such as metalwork, woodwork, and so forth. Female students have a different experience than male students do, as they are usually allocated to home economics, where they would engage in activities related to cooking, clothing, and textiles. Both students would have limited spatial experiences which exposed them to activities needing spatial skills such as rotating and viewing objects from different directions, but these spatial activities would not be consistent or specific in spatial training as the TD course. The students allocated to subjects within industrial arts and home economics are becoming very flexible, such that male and female students have the option to choose.

TD instruction

In the participating government secondary schools, students were taught an official national curriculum in which the TD was instructed for three 40-min sessions each week for 10 weeks by their teachers, while students not instructed in TD would attend their chosen subject options. During the first term (September to December) of the 2016–2017 academic year, the third-form students were taught TD for 10 weeks to develop their spatial visualization and spatial orientation skills. The teachers used various activities and strategies that included manipulatives, paper and pencil activities, chalkboard illustrations, and computer activities. During the research, the teachers were asked to use chalkboard, paper-and-pencil activities, and manipulatives. This restriction was employed to reduce those without computers being disadvantaged because some schools did not have or used computers within their instruction. The outline of topics for the 10 weeks of TD instruction is presented next.

In Week 1 (introduction to course), the students were introduced to various instruments (Module 1), concepts, and standards related to TD. Also discussed was the need for spatial visualization and spatial orientation skills in the national curriculum of Barbados, which follows the Caribbean Examination Council TD syllabus (Unit 1 Plane and Solid Geometry: Modules 1, 7, and 8).

In Weeks 2 and 3 (sketching of isometric and orthographic projections [Modules 7 and 8]), teachers explained the isometric projection angle of 30° and orthographic projection angles of 90° and 45° and standard layout. Manipulatives in the form of simple block objects were given to students to allow for concrete referencing while the teacher explained various concepts. The teachers sketched isometric and orthographic projections of these objects on the chalkboard while students copied on graph paper. Within the later weeks, students were given increasingly difficult objects, and the teachers reduced the frequency of assistance on exercises.

In Weeks 4–7 (isometric and orthographic drawing [Modules 7 and 8]), teachers were drawing the isometric and orthographic projections with drawing instruments on plain paper. Students were given objects to draw in isometric and orthographic projections and taught the standard layout of their drawings.

In Weeks 8–10 (object rotation), the concept of rotation of objects about vertical and horizontal planes was introduced. Students were taught to rotate objects about one axis with manipulatives. Teachers sketched and drew simple block objects in isometric projection, which students copied with manipulatives as references of the objects. Other activities involved students sketching and drawing the rotation of more complex blocks.

Data collection procedure

Within the first week of the school term, researchers administered a spatial test as the pretest to both experimental (TD) and control (no TD) groups. Students were asked to complete demographic information on their test answer sheets immediate before the test. The students were given 40 min to complete the test. After the pretest was administered, TD instruction occurred, in which students reproduced sketches or drawings of two- and three-dimensional figures each week for 10 weeks. The students in the control group followed their regular classes of sciences, home economics, and so forth. Students in the experimental group were presented with exercises each week that should be completed and submitted in the following week of class. At the end of these 10 weeks of instruction, the spatial test was administered as the posttest to both groups.

Data collection instrument

Spatial test battery

The Spatial test battery (STB) consisted of demographic information and two subtests. Demographic information regarding playing video games (yes or no), gender and name, was recorded by students on their test answer sheet. These subtests were modified versions of the MRT (Vandenberg & Kuse, 1978) used to measure spatial visualization ability, and the Object Perspective Test/SOT (Hegarty & Waller, 2004; Kozhevnikov & Hegarty, 2001) used to measure spatial orientation ability. Each of the multiple-choice scales consisted of 10 items selected from original tests because they were aligned with the TD curriculum. This curriculum involved tasks and activities that encouraged the development of different spatial abilities such as mental rotation of objects (spatial visualization) and mentally changing one's looking direction (spatial orientation).

Administered first was the 15-min MRT (see Figure 2), which involved mentally rotating three-dimensional block figures on the right to determine which two figures are similar to the criterion figure on the left. Students selected their responses (A, B, C, D) to each item on their test answer sheet. This original test of 24 items was previously examined for the estimate of reliability using Kuder–Richardson Formula 20 (KR 20) computed as greater than .80 by Sorby and Baartmans (2000).

Administered immediately after, was the 25-min SOT (see Figure 3) which involved two-dimensional figures being

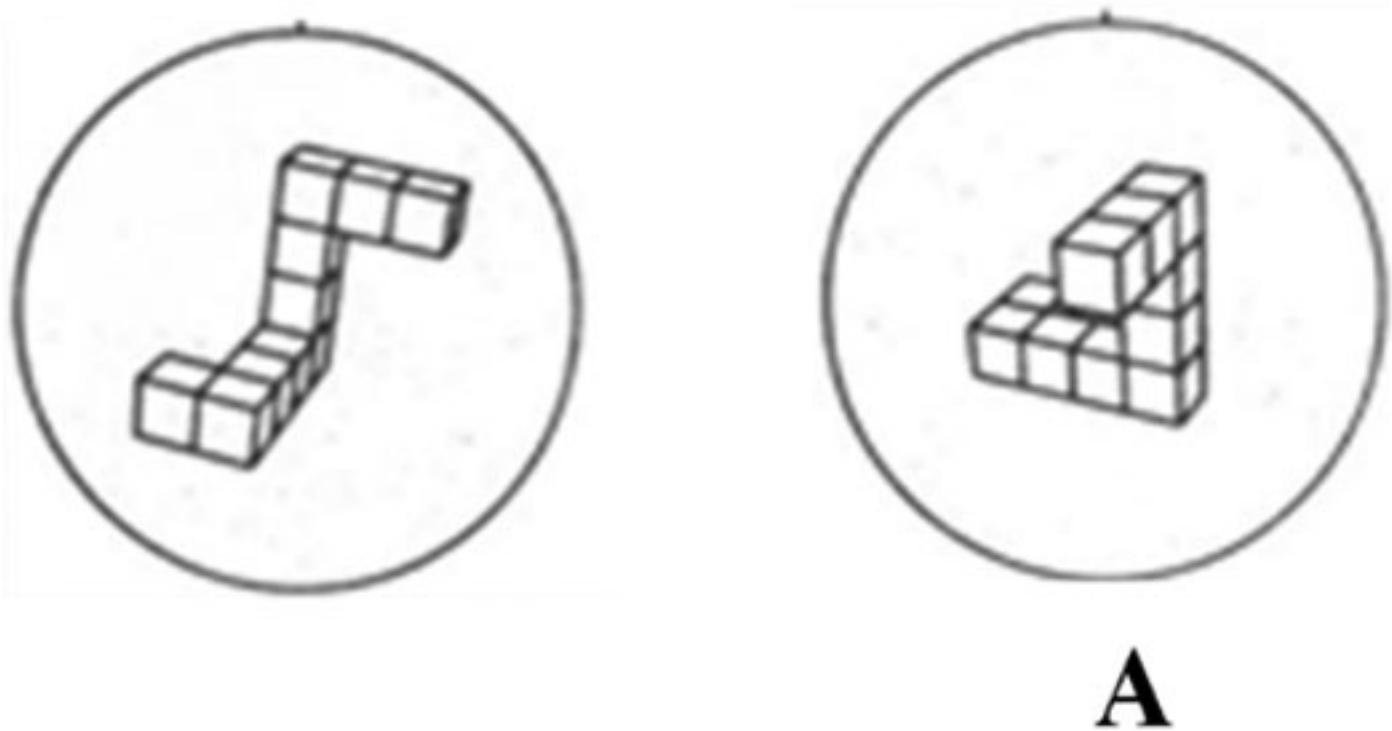

Figure 2. Example of Mental Rotations Test item.

viewed from various angles by mentally changing the looking directions. Students selected their responses (A, B, C, D) to each item on their test answer sheet. The original test of 12 items involved drawing an angled line that represented the looking direction. A comparison was performed between the drawn line and the correct looking angle to obtain a score. This test was previously examined for the estimate of reliability using a Cronbach's alpha computed as .79 by Hegarty and Waller (2004). We decided to modify the test by eliminating the drawn line representing the looking direction angle to accomplish age appropriateness and scoring accuracy.

The reliability and validity of the STB were examined in the present study, as four TD teachers evaluated the test items for alignment with the Barbadian curriculum, clarity of wording, age appropriateness, and level of the third-form students. After changes to the items, the instrument was piloted to 60 students in four secondary schools. The STB multiple-choice items were assumed to have equal difficulty respective to the subtests, the KR 20 (Nunnally & Bernstein, 1994) was used to estimate the reliability computed as .79 for the MRT and .70 for the SOT.

The STB analysis was based on two components namely spatial visualization and spatial orientation. Scores acquired from the mental rotation items comprised the visualization scores, and scores acquired from the spatial orientation items comprised the spatial orientation scores.

Data analysis

In this present study, a quantitative research methodology was used. Both descriptive and inferential statistics were used to determine the effects of a TD course, video games, gender, and type of school on third-form students' spatial ability namely spatial orientation, and spatial visualization (mental rotation). Any statistical analyses were performed using SPSS version 17. Assumption testing was conducted for violations of outliers (73) and missing data (116), which were removed resulting in 420 responses from a total of 609. Using Kolmogorov-Smirnov statistics for both subtests, a violation of normality occurred. However, inspection of the normal probability plots (Q-Q plots) suggests tenable normality. Stevens (1996) offered that violating this assumption was common in large samples and tests for analysis of variance were robust to overcome the violations. Assessment of a construct in social sciences may show the causal nature of the measured construct. A final violation was the homogeneity of variances for both subtests, and hence a more stringent significance level was set at .01 (Stevens, 1996; Tabachnick & Fidell, 2001). Nevertheless, both significance levels of .01 and .05 will be offered in the analyses.

Concerning descriptive statistics, the mean and standard deviation values of the STB pretest and posttest were calculated. A four-way analysis of covariance and Multiple Classification Analysis (MCA) was performed to obtain inferential statistics (Andrews, Morgan, Sonquist, & Klem, 1973; Retherford & Choe, 1993). Also, partial eta squared was performed to find the effect size of the results (Tabachnick & Fidell, 2001), and was applicable for factorial analyses of variances (Campo, Eckardt, Findley, Cardinale, & Shiyko, 2017). Cohen (1988) benchmarks of small (.01), moderate (.06), and large (.14) were used to interpret the effect size and show variance in means as opposed to practical or educational significance.

Imagine you are standing at the **car** and facing the **traffic light**.

Point to the **stop sign**.

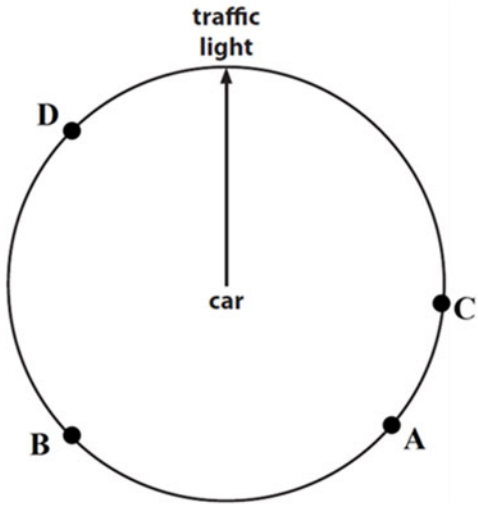

Figure 3. Example of Spatial Orientation Test item.

Table 2. Descriptive statistics for Mental Rotations Test and Spatial Orientation Test.

		Mental Rotations Test		Spatial Orientation Test	
Independent Variable	*n*	Pretest *M* (*SD*)	Posttest *M* (*SD*)	Pretest *M* (*SD*)	Posttest *M* (*SD*)
Technical drawing					
Yes	179	2.79 (1.09)	4.52 (0.99)	2.37 (0.89)	4.51 (0.97)
No	241	2.73 (1.02)	4.07 (0.83)	2.35 (0.83)	4.12 (0.95)
Video games					
Yes	380	2.80 (1.04)	4.31 (0.93)	2.37 (0.85)	4.38 (0.97)
No	40	2.38 (1.03)	3.83 (0.78)	2.25 (0.87)	3.45 (0.60)
Male	269	3.08 (0.97)	4.42 (0.94)	2.42 (0.78)	4.38 (0.95)
Female	151	2.18 (0.92)	3.98 (0.84)	2.25 (0.96)	4.13 (1.00)
Category 3 school	201	2.95 (1.09)	4.46 (0.98)	2.37 (0.83)	4.29 (0.95)
Category 4 school	219	2.58 (0.98)	4.08 (0.84)	2.34 (0.88)	4.29 (1.01)

Results

The purpose of this study was to investigate the effects of a TD course, video games, gender, and type of school on third-form students' spatial ability, namely spatial orientation and spatial visualization (mental rotation). A total of 420 secondary school students were assessed on the MRT and SOT and the descriptive statistics are given in Table 2.

Reliability was calculated using the KR 20, as this provided the internal consistency of the STB instruments. The KR 20 was calculated as .75, and .72 for the MRT and SOT, respectively. Salkind (2010) offered that .70 is a standard and sufficient value that is used in practice. In Table 2, there were increases in all pretest-to-posttest mean scores in both the MRT and SOT. Students exposed to TD and video games that were male and attended Category 3 type schools had overall higher mean scores in both the MRT and SOT. One can infer from these results that teachers must attempt to discover and manipulate those variables which affect students' spatial performance in spatial tasks and assessments. Success in teaching can be extended beyond the TD course, as spatial ability is important and relevant to several areas. The use of diverse and novel training methods occur in other courses to optimize the learning of spatial content. For the successful teaching of spatial concepts, teachers should consider the different effects of students' gender, prior experience with video games, and the type of school.

To determine whether the findings were statistically significant, inferential statistical analyses were performed. To be more precise, a four-way analysis of covariance was run to determine the main effect and combined interaction effects of the

independent variables (Stevens, 1996; Tabachnick & Fidell, 2001). An MCA was used as a post hoc analysis to further analyze statistical significance (Andrews, Morgan, Sonquist, & Klem, 1973; Retherford & Choe, 1993). The MCA was run to determine an increasing effect on third-form students' spatial visualization scores and spatial orientation scores.

The analysis of covariance summary tables (see Tables 3 and 4) present the MRT and SOT Posttest mean scores for a TD course, video games, gender, and type of school. Furthermore, these tables display the sum of squares, degrees of freedom, mean squares, F values, p values, and corresponding partial eta-squared values.

Main effect of independent variables on spatial visualization and orientation

An inspection of Table 3 revealed a significant main effect at the .05 level for the TD course on third-form students' spatial visualization, $F(1, 403) = 6.105$, $p = .014$, $\eta_p^2 = .015$. However, at the .01 level, there were no significant effects on the TD course, video games, gender, and type of school. Table 4 displays a statistical significant main effect at the .05 level for the TD course on spatial orientation, $F(1, 403) = 6.315$, $p = .012$, $\eta_p^2 = .015$. A significant main effect at the .01 level was revealed for video games, $F(1, 403) = 15.714$, $p < .0005$, $\eta_p^2 = .038$, but for neither gender nor type of school. All partial eta-squared values showed that the variance between posttest mean scores was small, indicating that the difference between group means was slight.

Two-way interaction effect of independent variables on spatial visualization and orientation

Table 3 showed a significant interaction at the .01 level for video games and gender, $F(1, 403) = 8.230$, $p = .004$, $\eta_p^2 = .020$, on third-form students' spatial visualization but not for all other possible combinations of the independent variables. Another significant interaction occurred at the .05 level for the TD course and gender, $F(1, 403) = 4.726$, $p = .030$, $\eta_p^2 = .012$, on spatial visualization. Graphical representation was further employed to show these significant disordinal interactions. Figure 4 indicates a significant interaction between TD and gender on spatial visualization. The male and female students exposed to a TD course had higher posttest mean scores on the MRT than those students not exposed, and boys achieved higher MRT posttest mean scores than girls did. Figure 5 displays a significant interaction between video games and gender on spatial visualization. The students responded differently to playing video games, with an inverse effect that was indicated by boys not playing video games had a higher MRT posttest mean scores than girls did. However, when these students played video games, the boys had declining scores and girls had increasing scores.

Table 4 indicated the only significant interaction effect occurred at the .05 level for video games and gender, $F(1, 403) = 4.123$, $p = .043$, $\eta_p^2 = .010$ on third-form students' spatial orientation. A graphical representation (see Figure 6) displayed this significant disordinal interaction of video games and gender on spatial orientation. Students who played video games had higher SOT posttest mean scores than those that did not; in favor of the female students.

Table 3. Analysis of covariance summary for MRT posttest according to independent variables.

Source of Variation	*SS*	*df*	*MS*	*F*	*p*	η_p^2
Covariate	68.514	1,403	68.514	125.794	.000	.238
Pretest for MRT	68.514	1,403	68.514	125.794	.000	.238
Technical drawing	3.325	1,403	3.325	6.105	.014	.015
VG	0.150	1,403	0.150	0.275	.601	.001
Gender	1.840	1,403	1.840	3.378	.067	.008
TS	0.029	1,403	0.029	0.054	.817	.000
Two-way interactions						
Technical drawing × VG	0.043	1,403	0.043	0.080	.778	.000
Technical drawing × Sex	2.574	1,403	2.574	4.726	.030	.012
Technical drawing × TS	1.594	1,403	1.594	2.927	.088	.007
VG × Gender	4.483	1,403	4.483	8.230	.004	.020
VG × TS	1.579	1,403	1.579	2.900	.089	.007
Gender × TS	0.700	1,403	0.700	1.285	.258	.003
Three-way interactions						
Technical drawing × VG × Gender	2.230	1,403	2.230	4.095	.044	.010
Technical drawing × VG × TS	1.814	1,403	1.814	3.331	.069	.008
Technical drawing × Gender × TS	0.008	1,403	0.008	0.015	.902	.000
VG × Gender × TS	0.551	1,403	0.551	1.011	.315	.003
Four-way interactions						
Technical drawing × VG × Gender × TS	1.115	1,403	1.115	2.047	.153	.005
Error	219.494	403	0.545			
Total	7997.000	420				
Corrected total	359.664	419				

Note. Independent variables are technical drawing, video games, sex, and type of school.
MRT = Mental Rotations Test; TS = type of school; VG = video games.

Table 4. Analysis of covariance summary for SOT posttest according to independent variables.

Source of Variation	*SS*	*df*	*MS*	*F*	*p*	η_p^2
Covariates	31.241	1,403	31.241	42.114	.000	.095
Pretest for SOT	31.241	1,403	31.241	42.114	.000	.095
Technical drawing	4.684	1,403	4.684	6.315	.012	.015
VG	11.657	1,403	11.657	15.714	.000	.038
Gender	0.027	1,403	0.027	0.037	.848	.000
TS	0.318	1,403	0.318	0.429	.513	.001
Two way interactions						
Technical drawing × VG	0.096	1,403	0.096	0.129	.719	.000
Technical drawing × Gender	0.073	1,403	0.073	0.098	.754	.000
Technical drawing × TS	0.781	1,403	0.781	1.053	.306	.003
VG × Gender	3.059	1,403	3.059	4.123	.043	.010
VG × TS	0.067	1,403	0.067	0.090	.765	.000
Gender × TS	0.809	1,403	0.809	1.091	.297	.003
Three-way interactions						
Technical drawing × VG × Gender	1.115	1,403	1.115	1.504	.221	.004
Technical drawing × VG × TS	1.527	1,403	1.527	2.058	.152	.005
Technical drawing × Gender × TS	0.589	1,403	0.589	0.793	.374	.002
VG × Gender × TS	0.210	1,403	0.210	0.283	.595	.001
Four-way interactions						
Technical drawing x VG × Gender × TS	0.014	1,403	0.014	0.019	.890	.000
Error	298.954	403	0.742			
Total	8132.000	420				
Corrected total	400.562	419				

Note. Independent variables are technical drawing, video games, gender, and type of school.
MRT = Mental Rotations Test; TS = type of school; VG = video games.

Three-way interaction effect of independent variables on spatial visualization and orientation

An inspection of Table 3 showed the only significant interaction effect at the .05 level for a TD course, video games,

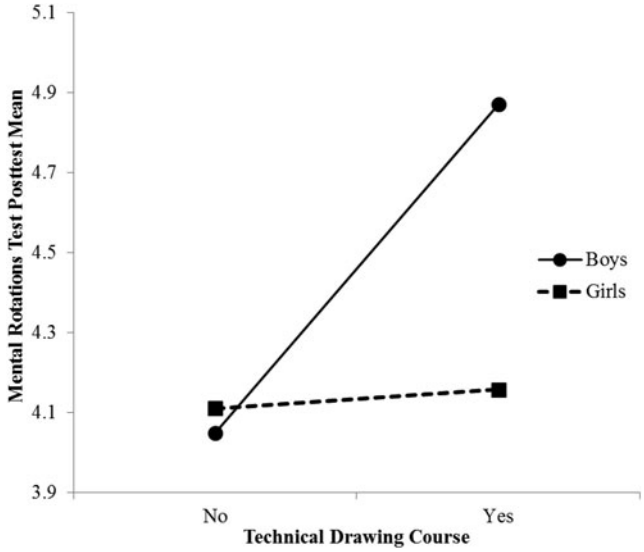

Figure 4. Interaction of technical drawing course and gender on spatial visualization of third-form students.

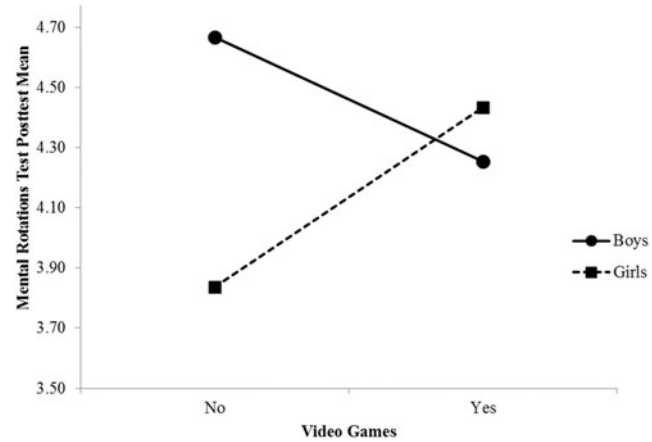

Figure 5. Interaction of video games and gender on spatial visualization of third-form students.

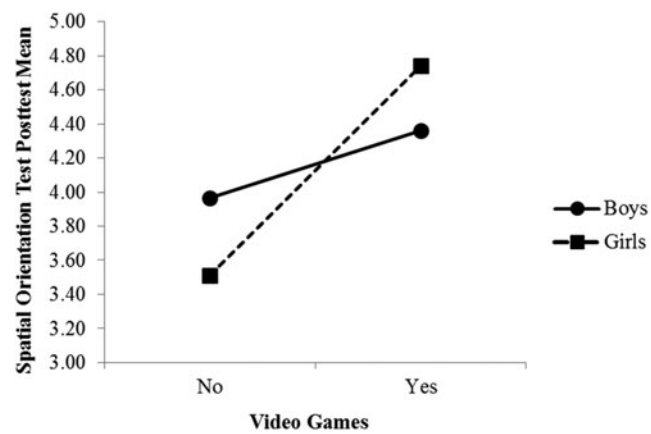

Figure 6. Interaction of video games and gender on spatial orientation of third-form students.

and gender, $F(1, 403) = 4.095$, $p = .044$, $\eta_p^2 = .010$, on third-form students' spatial visualization. The graphical representation (see Figure 7) illustrated this significant interaction, as students exposed to a TD course or video games had higher MRT posttest mean scores than did those that were not. Male students that were exposed to only the TD course had the highest overall posttest mean scores, with great disparity from female scores. However, those boys exposed to the TD course and video games combination had lower scores with less disparity between the girls' scores. This result suggested that male students performed better in the MRT when exposed to only a TD course, while female students may perform better when exposed to the combination of a TD course and video games. Table 4 indicated no significant interaction effects for all possible combinations of the independent variables on third-form students' spatial orientation.

Four-way interaction effect of independent variables on spatial visualization and orientation

Tables 3 and 4 displayed no statistically significant interaction effects of the independent variables on third-form students' spatial visualization and orientation. The results were not surprising, as the type of school had not reached significance in any of the analyses.

Discussion

The purpose of this study was to investigate the effects of a TD course, video games, gender, and type of school on the spatial ability of third form secondary school students. Spatial ability was examined in the context of two components, namely spatial visualization (mentation rotation) and spatial orientation (McGee, 1979; Risma, 2013). The literature has investigated the notion, and illustrated that spatial ability could be developed through activities and instruction similarly performed in a TD course (Alias et al., 2002; Basham & Kotrlik, 2008; Kayhan, 2005; Olkun, 2003; Sorby et al., 2013; Uttal, Meadow, et al., 2013; Uttal, Miller, & Newcombe, 2013; Verner & Merksamer, 2015). These investigations focused on effects of video and computer games (Cherney et al., 2014; De Lisi & Wolford, 2002; Feng et al., 2007; Gagnon, 1985; Gold et al., 2018; McClurg & Chaille, 1987; Peters et al., 1995; Spence & Feng, 2010; Terlecki & Newcombe, 2005), sex (Bodner & Guay, 1997; Hubona & Shirah, 2006; Linn & Petersen, 1985; Maeda & Yoon, 2013; McGee, 1979; Neuburger et al., 2011; O'Connor, 1943; Safhalter et al., 2013; Spencer-Smith et al., 2013; Tartre, 1990; Yazici, 2013), and different types of schools or learning environments (Guzel & Sener, 2009; Kayhan, 2005; Sharobeam, 2016).

Analysis of data in the present study showed increases from pretests to posttests within the levels, but there were small mean differences across the levels. The results indicated some statistical significant main and interaction effects on the spatial ability scores regarding the TD course, video games, and gender. Students' spatial ability scores seemed to be increased by the TD course that is comprised of spatial training and tasks. The TD students were engaged in the manipulation of concrete objects, sketching, and drawing of two- and three-dimensional shapes to aid the development

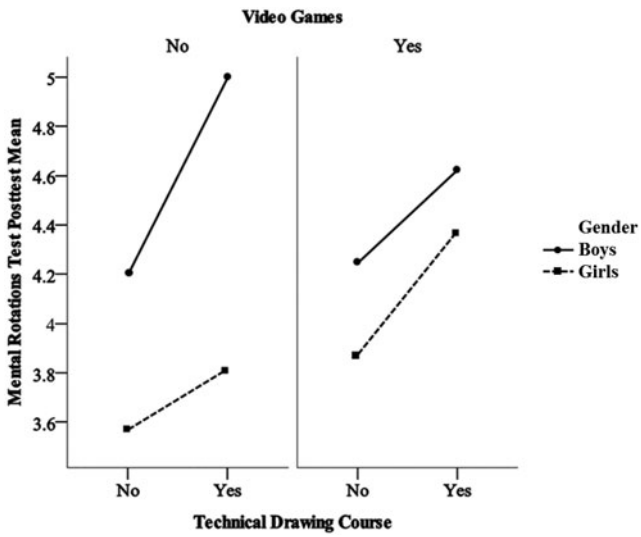

Figure 7. Interaction of technical drawing, video games, and sex on spatial visualization of third-form students.

of their spatial ability (spatial visualization and orientation skills). Uttal, Miller, and Newcombe (2013) specified that spatial training presented in a variety of methods was effective at increasing spatial skills. We found that TD involved several spatial training experiences using mental and physical tasks which progressively helped students to move from depending on physical objects to thinking about intangible objects. The national TD curriculum allowed for the transition from physical to intangible, in which students were exposed to physical manipulating (rotation and looking from different directions) solid, simple block objects for concrete referencing, which may have helped with internalizing concepts for meaningful learning. In later weeks, students were exposed to more mental manipulation of simple block drawings that became increasingly difficult in shape and how they were manipulated.

Although there was no exposure to TD instruction before entering the third form level, the students could have been exposed to spatial activities in other courses. It is plausible that activities in those courses could positively influence spatial development. However, these courses would not specifically and consistently target spatial training as compared with the TD course, which was indicated by the TD students having higher mean posttest scores. Results also indicated the presence of interactions that were not significant especially concerning the type of schools in Category 3 and Category 4. A possible reason for this lack of significant interaction regarding the type of schools could be that spatial experiences were encouraged within all schools due to the curriculum reform of Barbados (Curriculum, 2000). A result may be greater spatial experiences within all the schools that created an equalizing effect of reducing differences among the levels, especially in Category 3 and Category 4 schools.

The data display the overall benefits of the TD course and video games to a particular gender. A significant two-way interaction was found between TD and gender regarding spatial visualization (mental rotation). The results of increased student performance on the MRT when exposed to a TD course in favor of male students was consistent with Sorby, Casey, Veurink, and Dulaney (2013). Vandenberg and Kuse (1978) had similar results to our study favoring boys on the MRT, but they had not considered the effects of time limits on the test. Maeda and Yoon's (2013) meta-analysis findings suggested gender variation in mental rotation performance was attributed to the time limit on a test. Girls seem to have higher anxiety and need more time than boys when they attempt to complete a timed Purdue Spatial Visualization Tests (Visualization of Rotations); girls' anxiety increased when there was a strict time limit.

It should be noted that we had increased the time limit of the original MRT by an additional 5 min to reduce or eliminate the female students' anxiety, knowing it could be a factor. Allowing additional time could have impacted the results by providing adequate time to aid female students to complete all MRT items and have higher scores. However, Peters (2005) suggested that boys will outperform by girls on mental rotation tasks irrespective of time limits. We found significant two-way interactions between video games and gender regarding spatial visualization and spatial orientation. Our results were supported by De Castell, Larios, Jenson, and Smith (2015), in which the performance of participants increased and they outperformed novice players. In our study, the MRT performance of female students increased when they indicated that they played video games, but the male students had an inverse effect. In the two-way interaction for video games and gender regarding spatial orientation, both male and female students had increased performance on the SOT, in favor of the girls. Feng, Spence, and Pratt (2007) suggested that girls who have been trained with video games or have experience with video games seem to benefit more than boys, as the girls in our present study outperformed the male students. There were no further two-way interactions for all of the other possible combinations of the independent variables regarding spatial visualization and orientation. The only significant three-way interaction effect occurred among the TD course, video games, and gender on spatial visualization. The results showed that students exposed to a TD course benefitted and outperformed comparison groups, favoring boys. However, boys that were exposed to the TD course and played video games had lower performance on the MRT when compared with male nonplayers. The inverse was true, as female students that played video games had higher MRT posttest mean scores than female nonplayers did. Uttal, Meadow, et al. (2013) implied that video games were comparable to spatial training courses as they increased spatial skills and were beneficial to students who initially had poor spatial skills. However, they cautioned on the variation of time students play video games as it cannot be assumed is equal to all students. Cherney, Bersted, and Smetter (2014) offered that female students may benefit more from playing video games due to not having a ceiling effect, as in the case of boys. The female students in our study may have gained more from the TD course (formal training) and video games (informal training) because they could have greater possibility for development

due to having less prior opportunities with spatial activities than male students (Gold et al., 2018; Terlecki & Newcombe, 2005). Significance was not reached for a four-way interaction of a TD course, video games, gender, and type of school regarding spatial visualization and orientation.

The findings of this present study when compared with others, show that the present was more complex and examined greater factorial interactions. It is acknowledged that many educational researchers and practitioners who have examined the influences of training interventions, biological and social factors, and various environments related to spatial ability. The present study is novel, especially in the Caribbean context, as no known existing studies have explored the effects of this combination of variables on the spatial ability of secondary school students in Barbados and other Caribbean states. The present study provides a valuable contribution to spatial literature and learning environments research which examines the variables that affect spatial ability. Also, there is a further contribution to earlier research on examining the gender gap in spatial training. The use of spatial training programs such as TD, which used specific training methods to develop different spatial abilities, was considered. The literature has indicated that there is improvement of spatial performance through experiences with video game training in a classroom or informal play at home. As TD and video games have training merit on improving spatial performance, their interaction effect on spatial ability is considered. Gender-related differences may determine the effectiveness of programs, strategies, and technologies to be used within the teaching and learning environments. From this discussion, it is believed that there are implications for secondary school teachers, educational administrators, and other stakeholders. The first implication includes, all those influencing the educational context should consider interactions among TD, gender, video games, and other variables in the teaching of tasks related to spatial ability. In the second instance, teachers at the secondary level can use TD and video games to aid with optimal development of different spatial skills. However, planning lessons to incorporate video games or some innovative strategy to improve spatial ability is to be done with caution as particular gender may be disadvantaged. For the third implication, teachers in general but especially vocational teachers may need to pretest students to ascertain information before implementing novel tools and methodologies. The final implication is that educational stakeholders should ensure there are discussions regarding implementation and promotion to students, teachers, and parents about spatial training programs. Still, policymakers and school administrators must remain cognizant of newly implemented programs and courses that may affect student outcomes.

Limitations and direction for future research

Findings from this present study should be considered concerning the following limitations. First, the use of schools from Category 3 and Category 4 instead of the four categories, hence overgeneralization of findings is cautioned. Further research involving all categories is necessary to determine generalization and replicability as the type of school in Category 1, and Category 2 may produce different findings. Second, this study revealed the common unequal gender ratio of girls being represented in smaller proportions within the TD course. Future researchers should look to match the number of male and female students to reduce variances. Also, investigations are needed to determine why male students outnumber girls in TD courses. Third, there was a severe unequal ratio of approximately 1:10 between non–video game players versus video game players. This unequal ratio may occur due to 14-year-old students living in a technological era and being more likely to play video games than to not play video games. Any further explorations should equally match the number of players versus nonplayers, distinguish the types of video games students play and how these video games affect the development of spatial skills, and the amount of time spent playing the video games. Fourth, we acknowledge that students were instructed by different teachers employing their instructional strategies and tools, which may have an impact on the outcome of students. Hence, there were efforts to standardize the topics and instructional tools that focused on developing spatial visualization and orientation skills. However, there is a need for further examination into the effects of individual instructional strategies and tools used by teachers to develop the spatial ability of students.

Conclusion

Research has shown the positive effects of spatial training on the spatial ability of students and other aspects of their lives. In our present study, findings suggest that TD, video games, and gender affect third-form students' spatial ability, namely spatial visualization and orientation. Significant interactions were revealed although the effect sizes were small. The students may have encountered tasks that allowed them to use their spatial ability in other subjects but would not have engaged in specific and consistent spatial training to develop their spatial ability. Our findings can be used to show teachers, administrators, and external stakeholders that there are factors affecting students' spatial ability they should know. Furthermore, decisions must be purposeful on improving the educational context and practice for students. There are implications for technical and vocational teachers at the secondary school level, and other stakeholders involved in spatial training decisions. One of the implications is that individuals influencing educational contexts should consider interactions among TD, gender, video games, and other variables in the teaching of spatial ability concepts. A second implication is that teachers in general, but in this case at the secondary level, can use TD and video games to aid in optimally developing different spatial skills. However, planning lessons with video games or innovative instructional strategies to improve spatial ability is to be done with the gender of students considered. A third implication is that all teachers in general, but especially vocational

teachers, may need to pretest students to ascertain information before implementing novel tools and methodologies. A concluding implication is that educational stakeholders should frequently discuss implementing spatial training programs and promotion to students, teachers, and parents. Consequently, educational policymakers and school administrators must remain cognizant regarding the effects of implementing new programs or courses on student outcomes.

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